

LAB 1:

```
WITH ranked_films AS (  
  SELECT  
    c.name AS category_name,  
    f.title,  
    f.length,  
    DENSE_RANK() OVER (  
      PARTITION BY c.name  
      ORDER BY f.length DESC  
    ) AS film_rank  
  FROM film f  
  JOIN film_category fc ON f.film_id = fc.film_id  
  JOIN category c ON fc.category_id = c.category_id  
)  
SELECT  
  category_name,  
  title,  
  length,  
  film_rank  
FROM ranked_films  
WHERE film_rank <= 3  
ORDER BY category_name, film_rank;
```

	category_name character varying (25) 🔒	title character varying (255) 🔒	length smallint 🔒	film_rank bigint 🔒
1	Action	Worst Banger	185	1
2	Action	Darn Forrester	185	1
3	Action	Casualties Encino	179	2
4	Action	Quest Mussolini	177	3
5	Animation	Pond Seattle	185	1
6	Animation	Gangs Pride	185	1
7	Animation	Theory Mermaid	184	2
8	Animation	Sons Interview	184	2
9	Animation	Lawless Vision	181	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Searchers Wait	182	2
15	Comedy	Saturn Name	182	2

LAB 2:

WITH monthly_revenue AS (

SELECT

DATE_TRUNC('month', payment_date)::DATE AS month,

SUM(amount) AS current_revenue

FROM payment

GROUP BY DATE_TRUNC('month', payment_date)

),

lagged_revenue AS (

SELECT

month,

current_revenue,

LAG(current_revenue) OVER (ORDER BY month) AS previous_revenue

FROM monthly_revenue

)

```

SELECT
    month,
    current_revenue,
    previous_revenue,
    ROUND(
        ((current_revenue - previous_revenue) / previous_revenue) * 100.0,
        2
    ) AS mom_growth_pct
FROM lagged_revenue
ORDER BY month;

```

	month date	current_revenue numeric	previous_revenue numeric	mom_growth_pct numeric
1	2007-02-...	8351.84	[null]	[null]
2	2007-03-...	23886.56	8351.84	186.00
3	2007-04-...	28559.46	23886.56	19.56
4	2007-05-...	514.18	28559.46	-98.20

LAB 3:

```

SELECT
    payment_id,
    customer_id,
    payment_date,
    amount,
    SUM(amount) OVER (
        PARTITION BY customer_id
        ORDER BY payment_date
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT
    ) AS cumulative_clv
FROM payment
ORDER BY payment_date;

```

	payment_id [PK] integer	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_clv numeric
1	17793	416	2007-02-14 21:21:59.996577	2.99	2.99
2	18173	516	2007-02-14 21:23:39.996577	4.99	4.99
3	19399	239	2007-02-14 21:29:00.996577	4.99	4.99
4	18441	592	2007-02-14 21:41:12.996577	6.99	6.99
5	18698	49	2007-02-14 21:44:52.996577	0.99	0.99
6	19498	264	2007-02-14 21:44:53.996577	3.99	3.99
7	18686	46	2007-02-14 21:45:29.996577	4.99	4.99
8	18051	481	2007-02-14 22:03:35.996577	2.99	2.99
9	19036	139	2007-02-14 22:11:22.996577	2.99	2.99
10	18456	595	2007-02-14 22:16:01.996577	2.99	2.99
11	19239	191	2007-02-14 22:23:12.996577	2.99	2.99
12	18870	95	2007-02-14 22:41:17.996577	2.99	2.99
13	19265	197	2007-02-14 22:43:41.996577	2.99	2.99
14	18155	512	2007-02-14 22:57:03.996577	6.99	6.99
15	19293	210	2007-02-14 23:01:30.996577	2.99	2.99
16	18963	119	2007-02-14 23:05:16.996577	7.99	7.99

LAB 4 :

WITH customer_spending AS (

SELECT

co.country,

c.customer_id,

c.first_name,

c.last_name,

SUM(p.amount) AS total_spend

FROM customer c

JOIN payment p ON c.customer_id = p.customer_id

JOIN address a ON c.address_id = a.address_id

JOIN city ci ON a.city_id = ci.city_id

JOIN country co ON ci.country_id = co.country_id

GROUP BY co.country, c.customer_id, c.first_name, c.last_name

),

ranked_customers AS (

SELECT

country,

customer_id,

first_name,

last_name,

total_spend,

```

RANK() OVER (
    PARTITION BY country
    ORDER BY total_spend DESC
) AS rnk
FROM customer_spending
)
SELECT
    country,
    customer_id,
    first_name,
    last_name,
    total_spend
FROM ranked_customers
WHERE rnk = 1
ORDER BY country;

```

Showing rows: 1 to 108   Page No: 1 of 1    					
	country character varying (50) 	customer_id integer 	first_name character varying (45) 	last_name character varying (45) 	total_spend numeric 
1	Afghanistan	218	Vera	Mccoy	67.82
2	Algeria	176	June	Carroll	151.68
3	American Samoa	320	Anthony	Schwab	47.85
4	Angola	528	Claude	Herzog	93.80
5	Anguilla	381	Bobby	Boudreau	99.68
6	Argentina	560	Jordan	Archuleta	129.71
7	Armenia	41	Stephanie	Mitchell	118.75
8	Austria	266	Nora	Herrera	113.74
9	Azerbaijan	334	Raymond	Mcwhorter	108.75
10	Bahrain	590	Seth	Hannon	108.76
11	Bangladesh	21	Michelle	Clark	146.68
12	Belarus	144	Clara	Shaw	189.60
13	Bolivia	431	Joel	Francisco	92.78
14	Brazil	178	Marion	Snyder	194.61
15	Brunei	91	Lois	Butler	107.66
16	Bulgaria	540	Tyrone	Asher	112.76
17	Cambodia	516	Elmer	Mae	82.76
Total rows: 108		Query complete 00:00:00.312			CRLF L